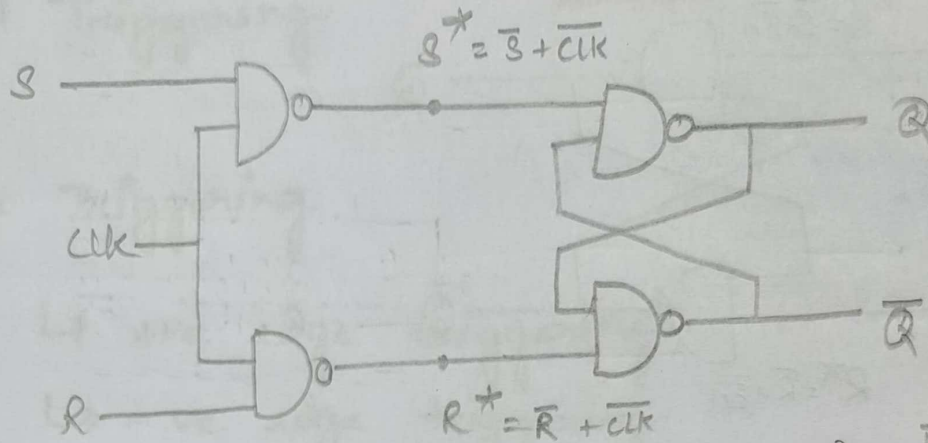


SR flip-flop:



Truth table
(S-R latch)

S	R	Q	$\overline{Q}$
0	0	not used	
0	1	1	0
1	0	0	1
1	1	memory	

Truth table:

CLK	S	R	Q	$\overline{Q}$
0	0	0	memory	
1	0	0	memory	
1	0	1	0	1
1	1	0	1	0
1	1	1	not used	

$$S^* = \overline{S} + \overline{CLK}$$

$$R^* = \overline{R} + \overline{CLK}$$

OR,

CLK	S	R	Q_{n+1}
0	x	x	Q_n
1	0	0	Q_n
1	0	1	0
1	1	0	1
1	1	1	Invalid

$(Q_n = \text{Present state})$
 $Q_{n+1} = \text{Next state}$

Characteristic Table :

(clk = 1)

Q_n	S	R	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	X
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	X

Q_n	S	R	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	X
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	X

$$Q_{n+1} = S + Q_n \bar{R}$$

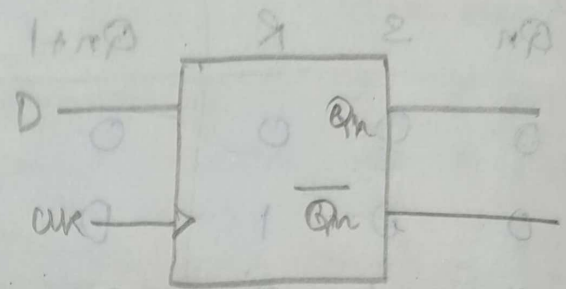
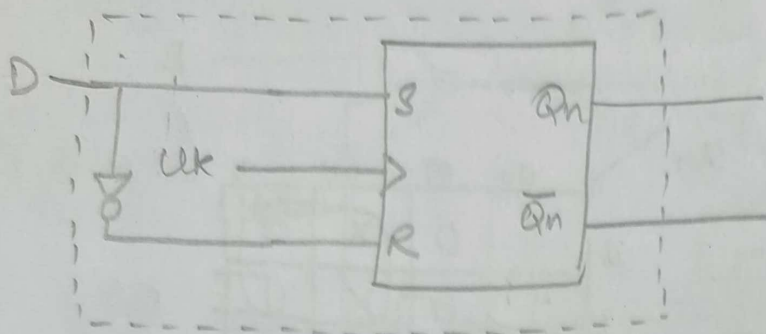
Excitation Table :

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

$$S = \overline{Q_n} Q_{n+1}$$

$$R = Q_n \bar{Q}_{n+1}$$

D Flip-flop:



D flip-flop

Truth table:

CLK	D	Q_{n+1}
0	x	Q_n
1	0	0
1	1	1

$$Q_{n+1} = D$$

Characteristic Table:

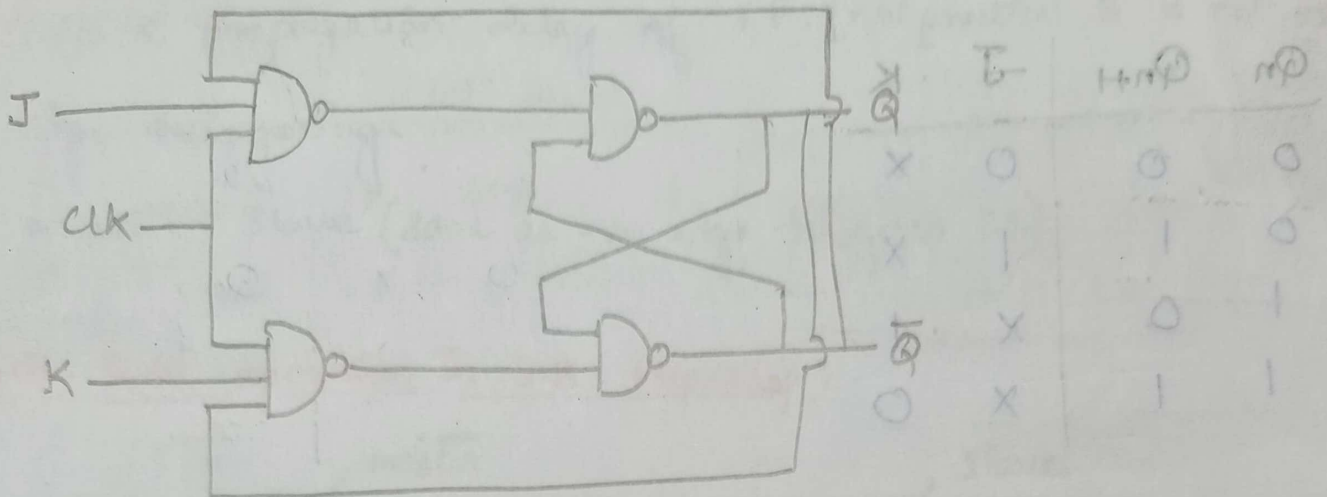
Q_n	D	Q_{n+1}
0	0	0
0	1	1
1	0	0
1	1	1

Excitation Table:

Q	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

JK Flip-Flop :

JK Flip-Flop



Truth Table :

CLK	J	K	Q _{n+1}
0	X	X	Q _n
1	0	0	Q _n
1	0	1	0
1	1	0	1
1	1	1	Q _n (toggle)

(Race condition)

when CLK=1, J=1, K=1

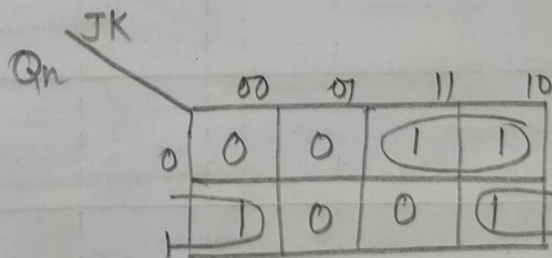
Assume Q=0 & Q-bar=1

Q = 0, 1, 0, 1, ...

Q-bar = 1, 0, 1, 0, ...

Characteristic Table :

Q _n	J	K	Q _{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0



$$Q_{n+1} = \overline{Q_n}J + Q_n\overline{K}$$

Excitation Table:

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

for J,

$Q_n \backslash Q_{n+1}$	0	1
0	0	1
1	X	X

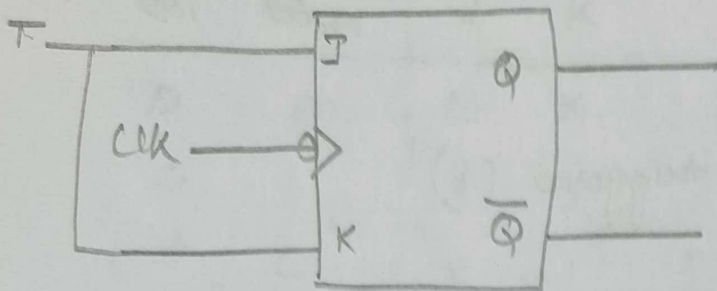
$$J = Q_{n+1}$$

for K,

$Q_n \backslash Q_{n+1}$	0	1
0	X	X
1	1	0

$$K = \overline{Q_{n+1}}$$

T flip-flop:



Truth Table:

clk	T	Q_{n+1}
0	x	Q_n
1	0	Q_n
1	1	$\bar{Q}_n$ (toggle)

memory

Characteristic Table:

Q_n	T	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

Excitation Table:

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0